Data Pipeline and Exploratory Analysis (Weeks 4–6)

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Court-IQ

Project Overview

Court-IQ forecasts NBA player performance using historical data and engineered features. In Weeks 4–6, we developed the data pipeline, performed exploratory analysis, and created a baseline model.

The target variables include MIN, PTS, REB, AST, STL, BLK, TOV, and PRA from the past five NBA seasons, with daily updates to ensure the data remains current. The ultimate goal of the application is to predict player performance before each game through a real-time, data-driven interface.

Week 4: Data Acquisition and Collection

Data was collected using the NBA_ Api library, primarily through the League Game Log endpoint. The dataset includes the last five NBA seasons with daily updates.

Advanced tracking and hustle metrics were also integrated to enhance traditional box score data. Custom Python scripts automate data retrieval and organization. All data used is publicly available for NBA statistics and contains no sensitive information.

Week 5: Data Preprocessing and Feature Engineering

We cleaned the data by fixing column names, correcting formats, and removing duplicate or incorrect records.

We created helpful features like recent game averages, rest days, home/away indicators, and tracking stats.

The data was then split into training, validation, and test sets in time order to prevent using future information.

Week 6: Exploratory Data Analysis

Exploratory analysis examined distributions, correlations, and performance trends across all target variables. Minutes played showed a strong relationship with scoring metrics, while defensive statistics displayed higher variability.

Rolling averages provided more stable signals than single-game values. A baseline regression model was implemented and evaluated using MAE and RMSE to establish initial predictive performance.

Conclusion

During Weeks 4–6, we successfully developed a structured data pipeline that transforms raw NBA statistics into a clean, feature-engineered dataset suitable for predictive modeling.

This phase established automated data collection, organized storage, feature engineering, time-based data splitting, exploratory analysis, and baseline model evaluation. The system now provides a strong foundation for developing and refining predictive models. All team members communicated effectively and worked together to successfully complete this phase of the project. We are now ready for the next phase of the project.

The Structure:

Data

- 10–15 seasons of NBA data
- ~500,000+ player-game records
- 40–80 engineered features
- Daily automated updates

Model Layer

- XG Boost / Random Forest
- Predicts PTS, PRA, REB, AST

- Confidence scoring included

Backend

- Fast API for real-time predictions
- Structured PostgreSQL database
- Secure REST endpoints

Frontend

- Stream lit dashboard
- Player search & analytics visualization

Deployment

- Cloud-ready (AWS/Azure)
- Docker containerized
- Scalable for SaaS expansion